

PROTOSTORIES - Spatial frames, NCU, Torun, Jul 2026 (#302703)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 5 authors.

Pre-registered on:
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1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

We test whether the size (element vs. complete) and spatial distance (proximate vs. displaced) of a referent affect how effectively it can be communicated through whole-body silent gesture, and which representational strategy (icon or index) communicators select to do so.

H1. Representations of complete spatial frames will have lower communicative effectiveness than representations of elements of spatial frames.

H2. There will be no difference in communicative effectiveness of the representations of proximate and displaced spatial frames, but they will call for different strategies:

(H2a) icons will be more likely to be used to communicate displaced targets than indexes; (H2b) indexes will be more likely to be used to communicate proximate targets than icons.

3) Describe the key dependent variable(s) specifying how they will be measured.

Two binary measures, one per model:

Communicative success (per trial): 1 if the guesser selects the correct target referent from the 16-image target response matrix, 0 otherwise.

Representation strategy (per production): the strategy used for the target referent, coded icon or index by two independent coders from 360° video.

Inter-rater agreement reported (Cohen's κ); disagreements resolved by discussion or a third coder.

4) How many and which conditions will participants be assigned to?

A fully within-dyad 2 (size: element vs. complete) $\times$ 2 (distance: proximate vs. displaced) design; every dyad completes all four conditions: (1) element-proximate, (2) element-displaced, (3) complete-proximate, (4) complete-displaced. Each dyad completes 16 trials (4 per condition); the communicator and guesser roles alternate between partners, and the first condition is counterbalanced across dyads.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

Bayesian hierarchical (Bernoulli, logit-link) models in brms with weakly informative priors, fit to all codeable trials.

'size', 'distance', and 'representation_strategy' are all sum-coded (± 0.5 ; e.g., icon = -0.5 , index = $+0.5$), so that lower-order terms are interpretable as main effects in the presence of the interaction and the intercept sits at the midpoint of the design.

Priors: Normal(0, 1.5) on fixed slopes, Normal(0, 2) on the intercept, Exponential(1) on RE standard deviations, LKJ(2) on RE correlations.

Model A: strategy (H2a/H2b)

representation_strategy ~ size*distance + (1 + size*distance | producer) + (1 + size*distance | guesser) + (1 | item)

Model B: success (H1 + H2 no-difference)

success ~ size*distance + representation_strategy + (1 + size*distance | producer) + (1 + size*distance | guesser) + (1 | item)

'item' enters with a random intercept only, as 'size' and 'distance' are constant within an item. If a model does not converge (target $\hat{R} \leq 1.01$ with no divergent transitions), we simplify the random-effects structure in a fixed order: drop the correlation terms, then the interaction slope, then the main-effect slopes (by-guesser before by-producer). As a robustness check, Model B is also fit without 'representation_strategy' to recover the total effect of size.

Hypotheses are evaluated against a region of practical equivalence (ROPE) of ± 0.18 on the log-odds scale (odds ratio 0.84–1.20; ± 0.30 reported as a sensitivity bound), using the 95% credible interval (CrI):

H1 is supported if the 'size' $\rightarrow$ success effect lies entirely below the ROPE (complete < element).

H2a/H2b are supported if the 'distance' $\rightarrow$ strategy effect lies entirely beyond the ROPE in the predicted direction (displaced $\rightarrow$ more icons); because strategy is binary, a single coefficient evaluates both.

The no-difference component of H2 is supported if the 'distance' $\rightarrow$ success effect lies entirely within the ROPE.

A CrI that straddles a ROPE boundary is treated as inconclusive.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We apply no outlier- or performance-based exclusions; all codeable trials enter the models. Trials lost to recording failure or uncodeable for strategy are excluded and their number reported. Any further exclusion is reported in the article.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will collect 20 dyads (40 participants), aiming for ≥ 20 productions/perceptions per unique item, cf. perception studies by Ćwiek et al. (2021, 2022, 2024), who report stable Bayesian estimation with as few as 7 data points per cell. After 20 dyads we fit the registered models. If the distance $\rightarrow$ success CrI straddles a ROPE boundary (equivalence unresolved), or any primary effect's 95% CrI is wider than 2 log-odds, we collect one further batch of 20 dyads (to 40) and re-assess once; we do not collect beyond 40.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We are preregistering the study around the same time we are starting data collection. Due to the availability of the infrastructure, the data collection was planned for the summer season, where convenient sampling from students and university employees, which we rely on, is limited and participant attrition is higher, hence, for pragmatic reasons, we had to start data collection as soon as possible. The design, the protocol, or any other details of the study have not changed between the start of the data collection and the preregistration.